



Estimating ICU bed capacity using discrete event simulation

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134

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Abstract

Purpose – The intensive care unit (ICU) in a hospital caters for critically ill patients. The number of the ICU beds has a direct impact on many aspects of hospital performance. Lack of the ICU beds may cause ambulance diversion and surgery cancellation, while an excess of ICU beds may cause a waste of resources. This paper aims to develop a discrete event simulation (DES) model to help the healthcare service providers determine the proper ICU bed capacity which strikes the balance between service level and cost effectiveness.

Design/methodology/approach – The DES model is developed to reflect the complex patient flow of the ICU system. Actual operational data, including emergency arrivals, elective arrivals and length of stay, are directly fed into the DES model to capture the variations in the system. The DES model is validated by open box test and black box test. The validated model is used to test two what-if scenarios which the healthcare service providers are interested in: the proper number of the ICU beds in service to meet the target rejection rate and the extra ICU beds in service needed to meet the demand growth.

Findings – A 12-month period of actual operational data was collected from an ICU department with 13 ICU beds in service. Comparison between the simulation results and the actual situation shows that the DES model accurately captures the variations in the system, and the DES model is flexible to simulate various what-if scenarios.

Originality/value – DES helps the healthcare service providers describe the current situation, and simulate the what-if scenarios for future planning.

Keywords Intensive care unit, Hospital beds, Bed capacity, Service levels, Rejection rate, Discrete event simulation

Paper type Research paper

1. Introduction

Intensive care units (ICU) in hospitals cater for critically ill patients who need immediate attention such as emergency cases and surgery recovery. Due to the critical patient conditions, the requests for the ICU beds have to be processed with no waiting time. Any delay could pose a significant threat to the patients' safety. Lack of the ICU beds may cause service level deteriorations including ambulance diversion, surgery cancellation and refusal of admission. On the other hand, excessive ICU beds may unnecessarily take up hospital budget, space and other valuable resources. Thus it is important for the healthcare service providers to determine the proper ICU bed capacity in order to strike the balance between service level and cost effectiveness. To make the situation more complex, the demand of the ICU beds is growing due to the aging and growing population, and the stochastic nature of emergency cases adds considerable fluctuation and variability to the demand of the ICU beds. All the above mentioned factors make the ICU beds planning a more challenging task.



The whole ICU system can be described as a queuing model with no waiting area (Mulligan, 1985), where patient arrival is accepted or rejected immediately. A typical queuing model is composed of three factors: arrival rate, service time and the number of servers. In the context of the ICU system, emergency and elective cases are regarded as the arrival rate, the length of stay of each case is regarded as the service time, and the number of the ICU beds is regarded as the number of servers. Once the distribution of the arrival rate, the length of stay and the number of servers are fixed, queue analysis can then be applied to calculate the relevant information such as rejection rate (Gorunescu *et al.*, 2002; Harper and Shahani, 2002; McManus *et al.*, 2004).

There are several problems of the above mentioned queuing model approach. First, the accuracy of queue analysis largely depends on how well variations (arrival rate and length of stay) in the ICU system are captured by the queuing model. However, it is difficult to match the variations with proper distributions. For instance, the arrival pattern of an ICU system is usually a mixture of emergency and elective cases. The emergency arrival varies according to time of the day, day of the week, and month of the year. The elective arrival is usually appointment based and can be considered as deterministic. It is difficult to find a proper distribution to match such a mixed arrival pattern. At the server's side, length of stay is highly skewed. Length of stay in the ICU is usually short because most patients in the ICU will be transferred to normal beds once stabilized. However, some complicated cases may stay in the ICU for a much longer period. Many distributions such as lognormal, gamma were applied to describe the distribution of length of stay, but the results did not match (Griffiths *et al.*, 2006). Some research works proposed queuing models with customized distribution. (Griffiths *et al.*, 2006) proposed a queuing model with negative exponentially distributed inter-arrivals and multiple service channels each having hyper-exponentially distributed service time. (Houdenhoven *et al.*, 2007) proposed a prediction model to estimate the length of stay for the queuing model.

Another problem of a queuing model is the difficulties to describe complex workflow. For instance, the ICU studied in this paper takes patients from two sources: emergency cases and elective surgeries, and the service providers are interested in performance indicators such as the number of cancelled surgeries. A typical queuing model may not meet such requirement.

Considering the limitations of a queuing model mentioned above, discrete event simulation (DES) is applied in this paper to model the patient flow of the ICU system. DES has been widely applied in many hospital sections including outpatient clinic (Harper and Gamlin, 2003; Klassen and Rohleder, 1996; Zhu *et al.*, 2009), emergency department (Connelly and Bair, 2004; Su and Shih, 2003; White *et al.*, 1992) etc. Simulation models of the ICU system are also introduced or developed in many research works (Cahill and Render, 1999; Costa *et al.*, 2003; Kim *et al.*, 1999; Ridge *et al.*, 1998). (Costa *et al.*, 2003) proposed a mathematical model to estimate the number of beds required using raw data. The mathematical model also provided the number of emergency patients transferred and the deferral rate of elective patients. (Kim *et al.*, 1999) applied queuing theory and computer simulation to estimate the proper utilization of the ICU beds, which took the trade-off between patient safety and cost effectiveness. (Ridge *et al.*, 1998) studied the relationship between the number of beds and rejection rate of the ICU requests due to lack of bed. Simulations showed a non-linear relationship between the two terms. (Cahill and Render, 1999) constructed a

simulation model to estimate the downstream and upstream resources needed by a fixed number of the ICU beds to avoid rejection or diversion of the ICU requests. (Krekeke *et al.*, 2004) compared various simulation techniques (mathematical modeling, Markov modeling, Monte Carlo simulation and DES) and their applications on the ICU system. Conclusion was drawn that DES provides more flexibility to describe the complexity of the ICU system than the other techniques.

The main contribution of this paper is to develop a generic approach, which estimates the ICU bed capacity using DES. Such an approach is composed of three steps: first, a DES model is constructed and validated to reflect the workflow of the ICU. Second, the bed capacity needed is estimated by the ICU by collecting the transaction data from ground and feeding into the model. Thirdly, different what-if scenarios the healthcare service provided are interested in are tested using the DES model and the results are used as references for decision making.

2. Method

Figure 1 illustrates the overall patient flow of the surgical ICU department of a Singapore government hospital studied in this paper. There are two sources of inflow: one source is the critically injured or ill patient from the emergency department, the other source comes from operating theatres. Part of the booked surgeries needs post-operative stay in the ICU. Both sources need immediate admission to the ICU once the requests are raised. A slight delay may affect patient safety severely and lead to irreversible consequence. Hence no waiting for the ICU beds or temporary buffer is allowed. If all the ICU beds are fully occupied, new emergency arrivals are overflowed to the ICU beds available in other departments or diverted to other hospitals. On the elective side, surgeries which need post-operative ICU stay have to be cancelled and rescheduled to a new appointment date and time. Both overflow and cancellation are considered as rejection. Length of stay in the ICU depends on patient's condition. There are four sources of outflow: discharge, transfer to high dependent bed, transfer to normal ward and death. The service providers are interested in the following five performance indicators: patient day, bed occupation rate, the number of overflowed cases, the number of cancelled cases and rejection rate. Patient day is defined as the

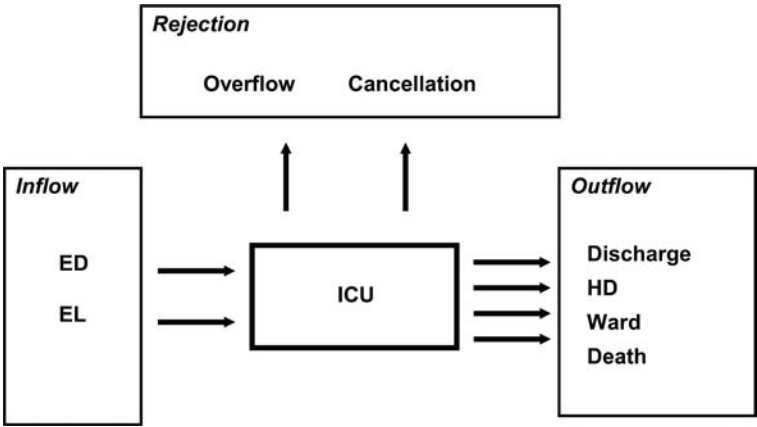


Figure 1.
Overall patient flow of the
surgical ICU

sum of length of stay of all admissions. Bed occupation rate is defined as the percentage of patient day over the total days provided by the ICU beds. Patient day and bed occupation rate describe the utilization of the ICU beds. The number of overflowed and cancelled cases is self-explanatory. Rejection rate is defined as the percentage of the rejected cases (overflow + cancellation) over the total arrivals.

Figure 2 illustrates the DES model based on the overall patient flow of the surgical ICU. Simul 8® 2008 professional is applied to construct the model. There are three flows in the DES model: The first flow is Emergency/Elective – Surgical ICU – Discharge/Transfer out/Death. This is the main flow when there are available ICU beds for emergency/elective arrivals. The second flow is Emergency – Surgical ICU – Other ICU. The flow occurs when no ICU bed is available when a new emergency case arrives. The unmet demand is overflowed to, if possible, other ICU or diverted to other hospitals. The third flow is Elective – Surgical ICU – Cancellation. The flow occurs when no ICU bed is available when a new elective case arrives. The elective case has to be cancelled and rescheduled for another slot in the waiting list.

In this paper, no distributions are assumed for both emergency and elective arrivals. Instead, actual operational data is fed into the simulation model directly. At the emergency side, the arrival time of patients who need immediate ICU admission is considered as the input. At the elective side, the appointment time of the elective patients to the ICU beds is regarded as the input. Length of stay of each individual patient is extracted from the actual operational data and paired with the arrival time or appointment time of the specific patient. The advantage of applying actual operational data directly is the possibility of reserving all the variations and seasonal factors existing in the ICU department.

One problem of using actual operational data in the simulation model is lack of length of stay information for the rejected patients, that is, emergency patients who are overflowed to other ICUs and elective patients whose appointments are cancelled. In this paper, length of stay of the rejected patients is sampled from the distribution of



Figure 2.
Discrete event simulation
model

length of stay of the actual operational data. Another problem of using actual operational data is that such data does not represent the future demand. In this paper, the future demand growth is simulated by inserting fictional patient arrivals into the actual operational data. Such arrival insertion is based on the following two rules: first, the demand growth is homogenously between emergency and elective cases, that is, the proportion between emergency and elective arrivals remains the same. Second, patient arrivals follow the same pattern as the original operational data. The variations of time of the day, day of the week and month of the year remain the same. The demand growth of each time period is in proportion to the original demand distribution. Length of stay of the inserted arrivals is sampled from the distribution of length of stay of the actual operational data.

3. Results and discussion

A two-step model validation procedure is applied in this paper to ensure that the simulation model accurately reflects the actual situation of the ICU department. The first step is open box validation. Patient flow of the model and parameter settings are presented to the service providers to confirm that the patient flow in the model is a valid mapping of the actual patient flow and the parameter settings are reasonable estimation of the actual practice. The second step is black box validation. Simulation results of the original scenario are measured by performance indicators and compared with the actual values from the operational data. Table I lists the results of black box validation, where the actual results are denoted by *A* and simulation results are denoted by *B*. The observation period is 12 months and there are 13 ICU beds currently in service. Both actual and simulation results are broken down by month. It is observed from Table I that obvious variations exist among different months. During the peak month, the ICU beds are more utilized with longer patient days and higher bed occupation rate. However, number of overflow and cancellation increases accordingly, which leads to a significantly higher rejection rate. It is also observed from Table I that the simulation model accurately captures the monthly variation of the actual situation. Simulation results of all five performance indicators are close to the actual results, which indicate that the simulation model is well calibrated. The simulation model can provide useful references during the evaluation of some what-if scenarios the service providers are interested in.

One what-if scenario the service providers are interested in is to decide the proper number of the ICU beds to mitigate the monthly variation. For instance, one of the requirements is to maintain a no more than 5 percent rejection rate in all months. Table I indicates that the requirement is only met in three months (January, September and December) in the original scenario. In the other months more ICU beds are needed in the peak months to satisfy the requirement. Table II and Figure 3 illustrate the simulation results of the relationship between the rejection rate and the number of the ICU beds in service. Three types of the ICU bed capacity (14, 15, and 16 ICU beds in service) were simulated in this study. It is observed from both Table II and Figure 3 that the rejection rate decreases significantly when there are more ICU beds in service. It is also observed from Figure 3 that the 5 percent rejection rate requirement is met in 7, 9, 12 months when the number of the ICU beds in service is 14, 15, 16 respectively. If the number of the ICU beds in service is fixed for all months, 16 ICU beds are needed to ensure that the 5 percent rejection rate requirement is met in all months. Table II also

Month	Patient days		Bed occupation rate (%)		Number of overflow		Number of cancellation		Rejection rate (%)	
	A	B	A	B	A	B	A	B	A	B
January	252	255	63	63	3	3	1	1	5	5
February	290	299	80	82	10	9	2	2	9	8
March	313	307	78	76	11	12	2	3	9	10
April	336	330	86	85	14	16	4	5	10	12
May	333	329	83	82	14	15	2	4	11	13
June	257	261	66	67	8	8	2	1	9	8
July	237	243	59	60	8	7	1	1	8	7
August	321	314	80	78	12	13	3	3	11	12
September	234	236	60	61	3	3	2	1	5	5
October	263	270	65	67	9	8	3	2	9	8
November	290	283	74	73	5	6	2	2	7	8
December	284	277	71	69	3	3	1	1	4	4
Total	3,410	3,404	72	72	100	103	25	25	9	9

Table I.
Comparison between
actual and simulation
results by monthly
breakdown

Month	ICU beds in service	Patient day	Bed occupation rate (%)	Number of overflow	Number of cancellation	Rejection rate (%)
January	14	263	61	1	0	1
	15	266	57	0	0	0
	16	266	54	0	0	0
February	14	303	77	7	2	6
	15	311	72	5	1	4
	16	319	68	3	0	2
March	14	318	73	9	1	7
	15	326	70	7	0	5
	16	335	68	4	0	3
April	14	342	81	13	3	9
	15	352	78	11	1	7
	16	360	75	8	0	5
May	14	338	78	13	2	11
	15	350	75	10	0	7
	16	358	72	7	0	5
June	14	270	64	6	0	5
	15	273	61	4	0	4
	16	279	58	1	0	1
July	14	246	57	5	1	5
	15	255	55	3	0	3
	16	259	52	0	0	0
August	14	328	76	10	1	8
	15	338	73	8	0	6
	16	343	69	6	0	5
September	14	244	58	1	0	1
	15	248	55	0	0	0
	16	248	52	0	0	0
October	14	278	64	5	1	5
	15	285	61	3	0	2
	16	289	58	1	0	1
November	14	291	69	4	1	5
	15	302	67	2	0	2
	16	306	64	0	0	0
December	14	287	66	1	0	1
	15	290	62	0	0	0
	16	290	58	0	0	0
Total	14	3,508	69	75	12	6
	15	5,473	66	53	2	4
	16	3,652	63	30	0	2

Table II.
Simulation results of the
performance indicators
under different ICU bed
settings

lists performance indicators other than rejection rate. It is observed that the adding more ICU beds does not necessarily increase patient day in idle months (January, September and December). Considering the existence of month variation, the practice of a fixed large number of the ICU beds in all months (16 ICU beds in this study) may cause waste of resources in idle and normal months. A more costs effective approach is to match the number of the ICU beds in service with the actual monthly volume. According to Table II and Figure 3, the original 13 ICU beds are sufficient in January, September and December, and 14 ICU beds are needed in June, July, October and

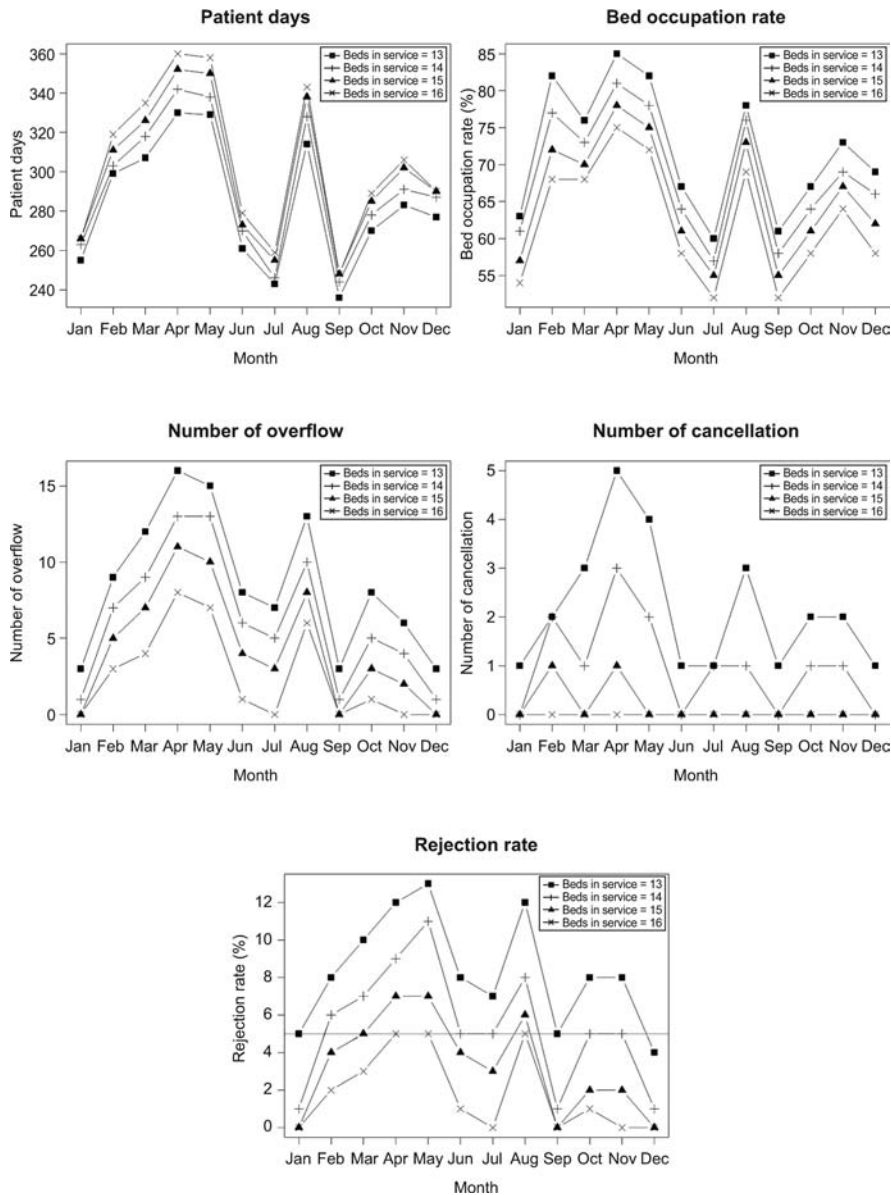


Figure 3.
Simulation results of
performance indicators
under different settings of
the ICU beds in service

November, 15 beds are needed in February and March, 16 beds are needed in April, May and August respectively to meet the 5 percent rejection rate.

Another scenario the service providers are interested in is the number of the ICU beds in service needed to handle the future demand growth. In this paper, two growth rates were simulated: 5 percent and 10 percent. The growth rate is assumed to be

homogenous between emergency and elective admissions. This assumption is based on the fact that the health care service providers in the ICU system are mostly interested in the overall demand growth and the source of the demand is relatively less significant. Table III illustrates the simulation results of the number of the ICU beds in service needed to meet the 5 percent rejection rate under the two suggested growth rate settings. It is observed that more ICU beds are needed to handle the growth. Averagely speaking, one extra ICU bed is needed when the growth rate is 5 percent, and two extra beds are needed when the growth rate is 10 percent.

The above results show how the DES model applied to the surgical ICU department in a Singapore government hospital. Necessary adjustments are needed when the DES model is applied to other hospitals. For instance, the flow of the DES model needs to match the workflow of the ICU in other hospitals. The what-if scenarios may also vary to reflect the interests of the service providers in other hospitals.

4. Conclusion

This paper developed a DES model to simulate the complex patient flow in a surgical ICU department of a Singapore government hospital. The DES model takes two sources of inflow: emergency and elective cases. The ICU beds are assigned on first come first serve basis. Once all the ICU beds are occupied, the incoming emergency

Month	Growth rate (%)	ICU beds in service	Patient day	Bed occupation rate (%)	Number of overflow	Number of cancellation	Rejection rate (%)
January	5	14	268	62	3	1	5
	10	15	282	61	3	1	4
February	5	15	324	77	5	2	5
	10	16	343	77	5	1	4
March	5	16	344	69	5	1	4
	10	17	360	68	5	2	4
April	5	17	374	73	7	2	5
	10	18	390	72	8	2	5
May	5	17	381	72	6	1	5
	10	18	401	72	5	2	4
June	5	15	285	63	4	1	4
	10	16	298	62	4	1	4
July	5	15	263	57	3	1	3
	10	16	275	55	4	1	4
August	5	17	356	68	6	1	5
	10	18	374	67	5	2	5
September	5	14	253	60	3	0	4
	10	15	262	58	3	1	5
October	5	15	302	65	3	1	2
	10	16	315	64	3	1	3
November	5	14	306	73	4	1	5
	10	15	320	71	4	1	5
December	5	13	286	71	4	1	5
	10	14	303	70	4	1	5

Table III.
Simulation results of the
performance indicators
under different growth
rates

cases will be overflowed to other departments or diverted to other hospitals and the elective cases will be cancelled. Actual operational data was collected and fed into the DES model to capture the variations in the system accurately. The calibrated model is used to test the what-if scenarios the healthcare service providers are interested in. Two what-if scenarios were tested in this paper: the proper number of the ICU beds in service to meet the target rejection rate and the extra ICU beds in service needed to meet the demand growth. Results show that the proposed DES model accurately describes the actual situation and is flexible enough to test the different what-if scenarios.

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